WORKSHEET · SCRATCH (GR 6–8) INTERMEDIATE: DATA & SIMULATIONS

Lesson 5: Decomposition and Abstraction

Module 2 | 45 minutes | Decomposition and refactoring | Grades 6–8 · ages 11–14

Name: _____ Class: _____ Date: _____

Practise what you learned today. Read each task carefully.

1. Key words

Match each word to what it means. Write the letter.

a. decomposition ___ A. Build small parts first, then combine.
b. abstraction ___ B. Improve code structure without changing behaviour.
c. refactor ___ C. Hiding details behind a simple name.
d. bottom-up ___ D. Breaking a problem into smaller parts.

2. Plan your code

Before you build it: A street: square (size) and roof (size) are defined. Draw a street of 3 houses: repeat 3 { square (50); roof (50) — or your own house block — then move 70 to the right without drawing }. Write or draw the blocks you will use.

3. Plan your code

Before you build it: Refactor: this Frog hops three times with the same 4 blocks copied three times. Make a hop block and call it 3 times, so the program uses 12 blocks or fewer and still ends at the same place (x = 60). Write or draw the blocks you will use.

4. Decompose “make a racing game” — which part is which?

Write each item in the right group: Player / Track / Rules.

a. steer with arrow keys → _____
b. draw the road edges → _____
c. lap counter → _____
d. speed up with the up arrow → _____
e. crash when touching the grass → _____
f. place the finish line → _____

5. Explain your thinking

Decompose a project you want to build into at least 4 procedures. Name each one and say what it does.

How did it go? Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

☐ I can decompose a problem into procedures.
☐ I can build bigger procedures from smaller ones.
☐ I can refactor repeated code.